

Birmjune Kim

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Education

Seoul National University (SNU), Seoul, Republic of Korea

· B.S. in Computer Science and Engineering

Mar 2024 - Present

· Kim Jaechul AI Class

Sep 2026 - Present

· Cumulative GPA: 4.24 / 4.3 · Major GPA: 4.27 / 4.3

Seoul Science High School, Seoul, Republic of Korea

Mar 2021 - Feb 2024

Research Interests

- **Efficient AI.** Making efficient AI models across various levels of the stack: inference-time methods, model compression and quantization, AI compilers, and hardware acceleration.
- **AI in the Physical World.** Putting those efficient models to work in the physical world (like robotics), where latency, memory, and power are real constraints.

Experience

Machine Learning Lab, Seoul National University

Dec 2025 - Present

Research Intern (Advisor: Prof. Hyun Oh Song)

Seoul, South Korea

- Researching a dynamic programming based dynamic decoding process for time series foundation models.

Projects

SNU AI Challenge 2026: Storyline Frame Ordering

Jul 2026 - Aug 2026

- Given a caption describing a storyline and four shuffled frames from that story, predict each frame's position in the correct temporal order.
- One of 12 finalist teams out of 206.

LLVM Optimization for a Custom Target Machine

May 2026 - Jun 2026

- An LLVM IR optimization pipeline for a custom target machine, with custom arithmetic, memory, and loop passes ahead of a provided backend.
- 6th of 24 teams in the course competition.

Image2Answer: 4-Option MCQ Solver

Aug 2025

- A vision-language pipeline that reads a question image and predicts the correct choice among the four options.
- Samsung Collegiate Programming Challenge (AI track): one of 41 finalists out of 1,483 entrants.

Afterglow: Event Photo Collection and Highlight Video

May 2025

- Collects event photos through a QR code and replays them as a highlight video.
- Built with a four-person team at the 3rd SKYST Joint Hackathon; Entrepreneur Award (1st in business idea).

Easy ARC-AGI Challenge Solver

Apr 2025 - Jun 2025

- Infers the rule behind a few example grid pairs and applies it to an unseen input, for grids up to 10×10 .
- Private score 63/100.

RISC-V CPU in Bluespec

Mar 2025 - Jun 2025

- RISC-V processors in Bluespec SystemVerilog, from a four-cycle non-pipelined CPU to three- and five-stage pipelines and a cached three-stage design.
- Control hazards handled with epochs; data hazards with a scoreboard and with forwarding.

Honors and Awards

One of 23 finalist teams (1,603 teams), NEXON Young Programmers Cup (NYPC) 2026, Master track	2026
One of 12 finalist teams (206 teams), SNU AI Challenge 2026	2026
One of 41 finalists (1,483 entrants), Samsung Collegiate Programming Challenge (AI track)	2025
8th of 275 pairs, Simon Marais Mathematics Competition (Pairs division, East)	2024
National Science & Technology Scholarship (Full Scholarship)	2024
Best Award, Graduation Thesis in Mathematics, Seoul Science High School	2023
Gold Medal, Korean Mathematical Olympiad (KMO), high school division	2022
Best Award, R&E Research Competition in Mathematics, Seoul Science High School	2022

Skills

Programming. Python, PyTorch, C / C++, Shell **Languages.** Korean (native), English